

WEEK 3: FINANCIAL DATA IMPORT, TRANSFORMATION & TIME INTELLIGENCE

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B.sc Data Science & AI

Introduction

Week 3 of the Power BI project focuses on importing, preparing, transforming, and analyzing **Shopify stock market data**. The main purpose of this project is to understand how Shopify's stock price and trading activity change over different periods of time.

Power BI was used to import the Shopify stock dataset and prepare it for financial analysis. A dedicated **Date Table** was created using DAX to support time-based calculations. A relationship was established between the Date Table and the Shopify stock data. Several DAX measures were then created to calculate stock prices, trading volume, daily returns, previous-year values, and year-to-date values.

The final outcome of the project is an interactive **Shopify Financial Performance Dashboard** containing KPI cards, charts, slicers, and time-based analysis.

Objectives of the Project

The main objectives of the Week 3 project are:

- Import Stock Data

Import the Shopify stock market dataset into Power BI Desktop and prepare it for analysis.

- Prepare Data Using Power Query

Check the imported data, verify the data types, and make sure the financial fields are suitable for calculations.

- Create a Dedicated Date Table

Create a separate Date Table using the `CALENDAR ()` DAX function.

- Create a Date Hierarchy

Create a custom hierarchy using:

Year → Quarter → Month → Day

This allows users to drill down from yearly analysis to daily analysis.

- Establish a Relationship

Create a **1-to-many active relationship** between the Date Table and Shopify stock data.

- Create DAX Measures

Create financial measures for stock prices, trading volume, daily return, previous close, and year-over-year comparison.

- Apply Time Intelligence

Use:

- `DATEADD()`
- `TOTALYTD()`
- `SAMEPERIODLASTYEAR()`

for time-based financial analysis.

Dataset Used

Dataset Name: Shopify Stock Data

The main dataset used in this project is the **Shopify Stock Data**.

The dataset contains historical Shopify stock information, including the date, opening price, highest price, lowest price, closing price, adjusted closing price, and trading volume.

- The **Date** column identifies the trading date.
- The **Open** column represents the opening price of the stock.
- The **High** column represents the highest price recorded during the trading period.
- The **Low** column represents the lowest price recorded during the trading period.
- The **Close** column represents the closing price of the stock.
- The **Adjusted Close** column contains the adjusted closing price.
- The **Volume** column represents the amount of stock traded during the period.

These fields provide the basic financial information required for creating the measures and visualizations in the dashboard.

Data Import and Preparation

The Shopify stock dataset was imported into **Power BI Desktop**.

After importing the dataset, the available columns were checked in Power BI Data View. The date and financial columns were reviewed to ensure that they could be used correctly for calculations and visualizations.

The main data table created from the dataset is called **Shopify_Stock**.

The table contains the stock market fields such as date, open, high, low, close, adjusted close, and volume.

The imported data was then prepared for further DAX calculations and time-based analysis.

	date	1.2 open	1.2 high	1.2 low	1.2 close
	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%
1	5/21/2015 12:00:00 AM -04:00	2.799999952	2.874000072	2.411000013	2.568000013
2	5/22/2015 12:00:00 AM -04:00	2.607000113	3.109999895	2.599999905	2.831000013
3	5/26/2015 12:00:00 AM -04:00	2.980000019	3.033999992	2.907999992	2.964999992
4	5/27/2015 12:00:00 AM -04:00	3.066999912	3.081000009	2.700000048	2.964999992
5	5/28/2015 12:00:00 AM -04:00	2.755000114	2.773999929	2.648000002	2.744999992
6	5/29/2015 12:00:00 AM -04:00	2.724999905	2.737999916	2.687000036	2.720000013
7	6/1/2015 12:00:00 AM -04:00	2.720000029	2.743999958	2.660000086	2.733000013
8	6/2/2015 12:00:00 AM -04:00	2.746999979	2.746999979	2.664999962	2.694999992
9	6/3/2015 12:00:00 AM -04:00	2.676000118	2.75	2.650000095	2.743999992
10	6/4/2015 12:00:00 AM -04:00	2.742000103	2.749000072	2.551000118	2.677999992
11	6/5/2015 12:00:00 AM -04:00	2.665999889	2.739000082	2.609999895	2.739000013
12	6/8/2015 12:00:00 AM -04:00	2.690000057	2.747999907	2.681999922	2.697999992
13	6/9/2015 12:00:00 AM -04:00	2.664000034	2.845999956	2.664000034	2.834000013
14	6/10/2015 12:00:00 AM -04:00	2.845000029	2.888000011	2.795000076	2.809999992
15	6/11/2015 12:00:00 AM -04:00	2.848999977	3.072999954	2.848999977	3.072999992
16	6/12/2015 12:00:00 AM -04:00	3.131999969	3.369999886	3.125	3.362999992
17	6/15/2015 12:00:00 AM -04:00	3.397000074	3.812000036	3.292000055	3.549999992
18	6/16/2015 12:00:00 AM -04:00	3.605999947	3.837899923	3.584000111	3.811000013

Shopify_Stock Table

The **Shopify_Stock** table is the main stock data table used in the project.

It contains the historical stock observations that are analyzed in the dashboard.

- The **date** field identifies the date associated with each stock record.
- The **open** field contains the opening stock price.
- The **high** field contains the highest price recorded.
- The **low** field contains the lowest price recorded.
- The **close** field contains the closing stock price.
- The **adj_close** field contains the adjusted closing price.
- The **volume** field contains the trading volume.

The Shopify_Stock table acts as the main source for the financial measures and visualizations.

date	open	high	low	close	adj_close	volume
5/21/2015 12:00:00 AM	2.79999995231628	2.87400007247925	2.41100001335144	2.56800007820129	2.56800007820129	123039000
5/22/2015 12:00:00 AM	2.60700011253357	3.10999989509583	2.59999990463257	2.83100008964539	2.83100008964539	28412000
5/26/2015 12:00:00 AM	2.98000001907349	3.03399991989136	2.9079999237061	2.96499991416931	2.96499991416931	8202000
5/27/2015 12:00:00 AM	3.06699991226196	3.08100008964539	2.70000004768372	2.75	2.75	7976000
5/28/2015 12:00:00 AM	2.75500011444092	2.7739999294281	2.64800000190735	2.74499988555908	2.74499988555908	4000000
5/29/2015 12:00:00 AM	2.72499990463257	2.73799991607666	2.68700003623962	2.72000002861023	2.72000002861023	2554000
6/1/2015 12:00:00 AM	2.72000002861023	2.74399995803833	2.66000008583069	2.73300004005432	2.73300004005432	2569000
6/2/2015 12:00:00 AM	2.74699997901917	2.74699997901917	2.66499996185303	2.6949999332428	2.6949999332428	3798000
6/3/2015 12:00:00 AM	2.67600011825562	2.75	2.65000009536743	2.74399995803833	2.74399995803833	3024000
6/4/2015 12:00:00 AM	2.74200010299683	2.74900007247925	2.55100011825562	2.67799997329712	2.67799997329712	3226000
6/5/2015 12:00:00 AM	2.66599988937378	2.73900008201599	2.60999989509583	2.73900008201599	2.73900008201599	1294000
6/8/2015 12:00:00 AM	2.69000005722046	2.74799990653992	2.68199992179871	2.69799995422363	2.69799995422363	2809000
6/9/2015 12:00:00 AM	2.66400003433228	2.84599995613098	2.66400003433228	2.83400011062622	2.83400011062622	3375000
6/10/2015 12:00:00 AM	2.84500002861023	2.88800001144409	2.79500007629395	2.80999994277954	2.80999994277954	2734000
6/11/2015 12:00:00 AM	2.84899997711182	3.07299995422363	2.84899997711182	3.07299995422363	3.07299995422363	4606000
6/12/2015 12:00:00 AM	3.13199996948242	3.36999988555908	3.125	3.36299991607666	3.36299991607666	8474000
6/15/2015 12:00:00 AM	3.3970000743866	3.81200003623962	3.29200005531311	3.54999995231628	3.54999995231628	16943000
6/16/2015 12:00:00 AM	3.60599994659424	3.83789992332458	3.58400011062622	3.81100010871887	3.81100010871887	11095000
6/17/2015 12:00:00 AM	3.96300005912781	4.21299982070923	3.80100011825562	3.86299991607666	3.86299991607666	15952000
6/18/2015 12:00:00 AM	3.84200000762939	3.88899993896484	3.4326000213623	3.5550000667572	3.5550000667572	11482000

Creating the Date Table

A separate **Date Table** was created to support time-based analysis.

The Date Table was created using the DAX `CALENDAR()` function.

The formula used was:

```
Date Table =  
CALENDAR(  
    MIN('Shopify_Stock'[date]),  
    MAX('Shopify_Stock'[date])  
)
```

The `MIN()` function identifies the earliest date available in the Shopify_Stock table.

The `MAX()` function identifies the latest date available in the `Shopify_Stock` table.

The `CALENDAR()` function creates a continuous series of dates between the minimum and maximum dates.

Creating a dedicated Date Table is important because it provides a proper date dimension for time-intelligence calculations.

1 Date Table = CALENDAR(MIN('Shopify_Stock'[date]),MAX(Shopify_Stock[date]))						
Date	Year	Quarter	Month	Month Number	Day	
5/21/2015 12:00:00 AM	2015	Q2	May	5	21	
5/22/2015 12:00:00 AM	2015	Q2	May	5	22	
5/23/2015 12:00:00 AM	2015	Q2	May	5	23	
5/24/2015 12:00:00 AM	2015	Q2	May	5	24	
5/25/2015 12:00:00 AM	2015	Q2	May	5	25	
5/26/2015 12:00:00 AM	2015	Q2	May	5	26	
5/27/2015 12:00:00 AM	2015	Q2	May	5	27	
5/28/2015 12:00:00 AM	2015	Q2	May	5	28	
5/29/2015 12:00:00 AM	2015	Q2	May	5	29	
5/30/2015 12:00:00 AM	2015	Q2	May	5	30	
5/31/2015 12:00:00 AM	2015	Q2	May	5	31	
6/1/2015 12:00:00 AM	2015	Q2	Jun	6	1	
6/2/2015 12:00:00 AM	2015	Q2	Jun	6	2	
6/3/2015 12:00:00 AM	2015	Q2	Jun	6	3	
6/4/2015 12:00:00 AM	2015	Q2	Jun	6	4	
6/5/2015 12:00:00 AM	2015	Q2	Jun	6	5	

Date Table Columns

After creating the Date Table, additional calculated columns were created for different time periods.

The Date Table contains the Date, Day, Month, Month Number, Quarter, and Year fields.

These columns are used for filtering, grouping, sorting, drill-down analysis, and time-intelligence calculations.

Date Column

The **Date** column contains the continuous sequence of dates created using the `CALENDAR()` function.

This column is the most important field in the Date Table because it is used to establish the relationship with the date field in the `Shopify_Stock` table.

Day Column

The Day column extracts the day number from the Date column.

The DAX formula used is:

```
Day =  
DAY('Date Table'[Date])
```

This column is useful for detailed daily analysis.

Month Column

The Month column extracts the month name from the Date column.

The DAX formula used is:

```
Month =  
FORMAT('Date Table'[Date], "MMM")
```

The "MMM" format displays the month as an abbreviated name.

Month Number Column

The Month Number column extracts the numerical month from the Date column.

The DAX formula used is:

```
Month Number =  
MONTH('Date Table'[Date])
```

The Month Number is useful for sorting the Month column in chronological order.

Quarter Column

The Quarter column identifies the quarter associated with each date.

The DAX formula used is:

```
Quarter =  
"Q" & FORMAT('Date Table'[Date], "Q")
```

The result produces values such as **Q1, Q2, Q3, and Q4**.

This column is used to analyze stock trading activity across different quarters.

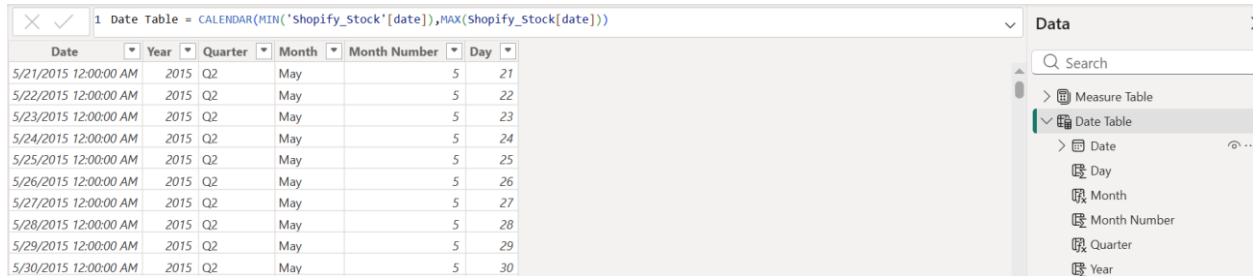
Year Column

The Year column extracts the year from the Date column.

The DAX formula used is:

```
Year =  
YEAR('Date Table'[Date])
```

The Year column is used for yearly analysis and is also the first level of the custom date hierarchy.



The screenshot shows a Power BI interface. On the left, a data table is displayed with columns: Date, Year, Quarter, Month, Month Number, and Day. The data represents dates in May 2015. On the right, the 'Data' pane shows a hierarchy: Measure Table, Date Table, Date, and then Day, Month, Month Number, Quarter, and Year.

Date	Year	Quarter	Month	Month Number	Day
5/21/2015 12:00:00 AM	2015	Q2	May	5	21
5/22/2015 12:00:00 AM	2015	Q2	May	5	22
5/23/2015 12:00:00 AM	2015	Q2	May	5	23
5/24/2015 12:00:00 AM	2015	Q2	May	5	24
5/25/2015 12:00:00 AM	2015	Q2	May	5	25
5/26/2015 12:00:00 AM	2015	Q2	May	5	26
5/27/2015 12:00:00 AM	2015	Q2	May	5	27
5/28/2015 12:00:00 AM	2015	Q2	May	5	28
5/29/2015 12:00:00 AM	2015	Q2	May	5	29
5/30/2015 12:00:00 AM	2015	Q2	May	5	30

Creating the Year Hierarchy

A custom **Year Hierarchy** was created in the Date Table.

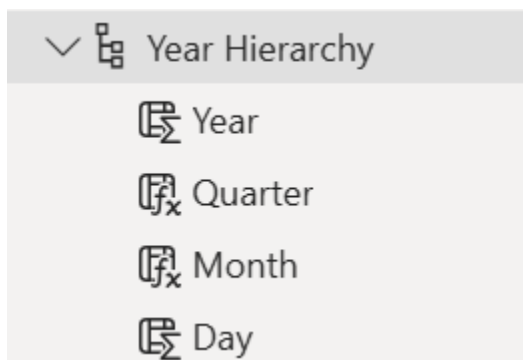
The hierarchy contains four levels:

Year → Quarter → Month → Day

This hierarchy allows users to drill down from a higher level of time analysis to a more detailed level.

For example, the analysis can begin with a particular year, then move to a quarter, then a month, and finally a specific day.

This provides a convenient way to analyze stock performance at different levels of time.



Relationship Between Date Table and Shopify_Stock

A relationship was created between the **Date Table** and the **Shopify_Stock** table.

The relationship connects:

Date Table[Date]

with

Shopify_Stock[date]

The relationship is a **one-to-many (1:*)** active relationship.

The Date Table is on the **one side**, while Shopify_Stock is on the **many side**.

The relationship can be represented as:

Date Table (1) → Shopify_Stock (*)

The Date Table contains unique dates, while the Shopify_Stock table contains the financial records associated with those dates.

New relationship

Select tables and columns that are related.

From table

Date Table

Date	Day	Month	Month Number	Quarter	Year
5/21/2015 12:...	21	May	5	Q2	2015
5/22/2015 12:...	22	May	5	Q2	2015
5/23/2015 12:...	23	May	5	Q2	2015

To table

Shopify_Stock

adj_close	close	date	date - Copy	high	low	open
2.5680000782...	2.5680000782...	5/21/2015 12:...	Thursday, Ma...	2.8740000724...	2.4110000133...	2.79999...
2.8310000896...	2.8310000896...	5/22/2015 12:...	Friday, May 2...	3.1099998950...	2.5999999046...	2.607000...
2.9649999141...	2.9649999141...	5/26/2015 12:...	Tuesday, May...	3.0339999198...	2.9079999923...	2.980000...

Cardinality

One to many (1:*)

Cross-filter direction

Single

☒ Make this relationship active

☐ Apply security filter in both directions

Save

Cancel

Measure Table

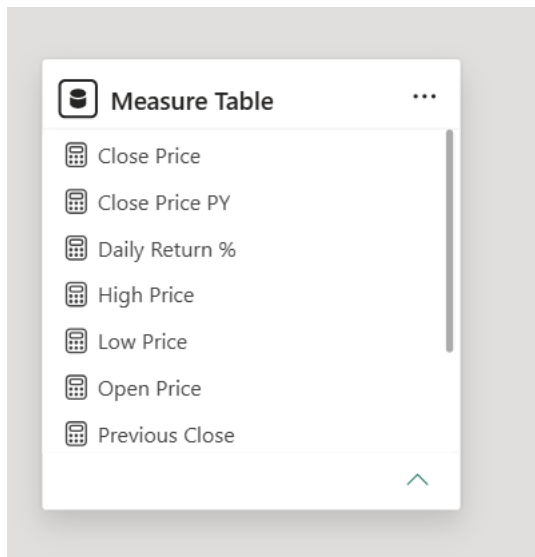
A separate **Measure Table** was created to store and organize the DAX measures.

The Measure Table contains the financial calculations required for the dashboard.

The measures created are:

- Close Price
- Close Price PY
- Daily Return %
- High Price
- Low Price
- Open Price
- Previous Close
- Trading Volume
- Trading Volume YTD
- Year-over-Year Percentage

Creating a separate Measure Table makes the Power BI model more organized and makes it easier to find and manage the DAX calculations.



Close Price Measure

The Close Price measure calculates the average closing price of Shopify stock.

The DAX formula used is:

```
Close Price =  
AVERAGE('Shopify_Stock'[close])
```

The `AVERAGE()` function calculates the average value of the closing price.

This measure is used to display the closing price based on the selected time period.

It is also used as a base measure for other calculations such as Close Price PY and Year-over-Year Percentage.

Close Price PY Measure

The **Close Price PY** measure calculates the closing price for the corresponding period in the previous year.

The DAX formula used is:

```
Close Price PY =  
CALCULATE(  
    [Close Price],  
    SAMEPERIODLASTYEAR('Date Table'[Date])  
)
```

The `SAMEPERIODLASTYEAR()` function changes the current date context to the corresponding period from the previous year.

This measure is useful for comparing the current closing price with the previous year's closing price.

Daily Return % Measure

The **Daily Return %** measure calculates the percentage change between the current closing price and the previous closing price.

The DAX formula used is:

```
Daily Return % =  
DIVIDE(  
    MAX('Shopify_Stock'[close]) - [Previous Close],  
    [Previous Close],  
    0  
)
```

The current closing price is compared with the previous closing price.

The `DIVIDE()` function calculates the percentage change.

The final 0 is used as an alternative result if the division cannot be performed.

This measure helps analyze the daily movement of Shopify stock.

High Price Measure

The High Price measure identifies the highest stock price within the selected period.

The DAX formula used is:

```
High Price =  
MAX('Shopify_Stock'[high])
```

The `MAX()` function returns the highest value from the High column.

This measure is used in the dashboard to display the maximum stock price.

Low Price Measure

The Low Price measure identifies the minimum stock price within the selected period.

The DAX formula used is:

```
Low Price =  
MIN('Shopify_Stock'[low])
```

The `MIN()` function returns the lowest value from the Low column.

This measure is used to display the minimum stock price in the dashboard.

Open Price Measure

The Open Price measure calculates the average opening price.

The DAX formula used is:

```
Open Price =  
AVERAGE('Shopify_Stock'[Open])
```

This measure provides the average opening stock price for the selected period.

It is displayed as one of the important KPI indicators in the dashboard.

Previous Close Measure

The Previous Close measure retrieves the closing price from the previous day.

The DAX formula used is:

```
Previous Close =  
CALCULATE(  
    MAX('Shopify_Stock'[close]),  
    DATEADD('Date Table'[Date], -1, DAY)  
)
```

The `DATEADD()` function shifts the current date context backward by one day.

The Previous Close measure is important because it is used in the calculation of the Daily Return % measure.

Trading Volume Measure

The Trading Volume measure calculates the total number of shares traded.

The DAX formula used is:

```
Trading Volume =  
SUM('Shopify_Stock'[volume])
```

The `SUM()` function adds the volume values for the selected period.

This measure is used in the trading-volume charts and helps analyze the level of trading activity.

Trading Volume YTD Measure

The Trading Volume YTD measure calculates the cumulative trading volume from the beginning of the year to the selected date.

The DAX formula used is:

```
Trading Volume YTD =  
TOTALYTD(  
    [Trading Volume],  
    'Date Table'[Date]  
)
```

The `TOTALYTD()` function performs a year-to-date calculation.

This measure provides cumulative trading activity for the current year and is used as a KPI in the dashboard.

Year-over-Year Percentage Measure

The Year-over-Year Percentage measure compares the current closing price with the previous year's closing price.

The DAX formula used is:

```
Year-over-Year-Percentage =  
DIVIDE(  
    [Close Price] - [Close Price PY],  
    [Close Price PY],  
    0  
)
```

This measure calculates the percentage difference between the current closing price and the corresponding previous-year closing price.

Time-Intelligence Functions

Time intelligence is used for time-based financial analysis in the project.

- **SAMEPERIODLASTYEAR()**

Retrieves the same period from the previous year for comparing closing prices.

- **DATEADD()**

Shifts the date context by **-1 day** to find the previous closing price.

- **TOTALYTD()**

Calculates the cumulative trading volume from the beginning of the year up to the selected date.

These functions help perform effective time-based analysis.

Dashboard Overview

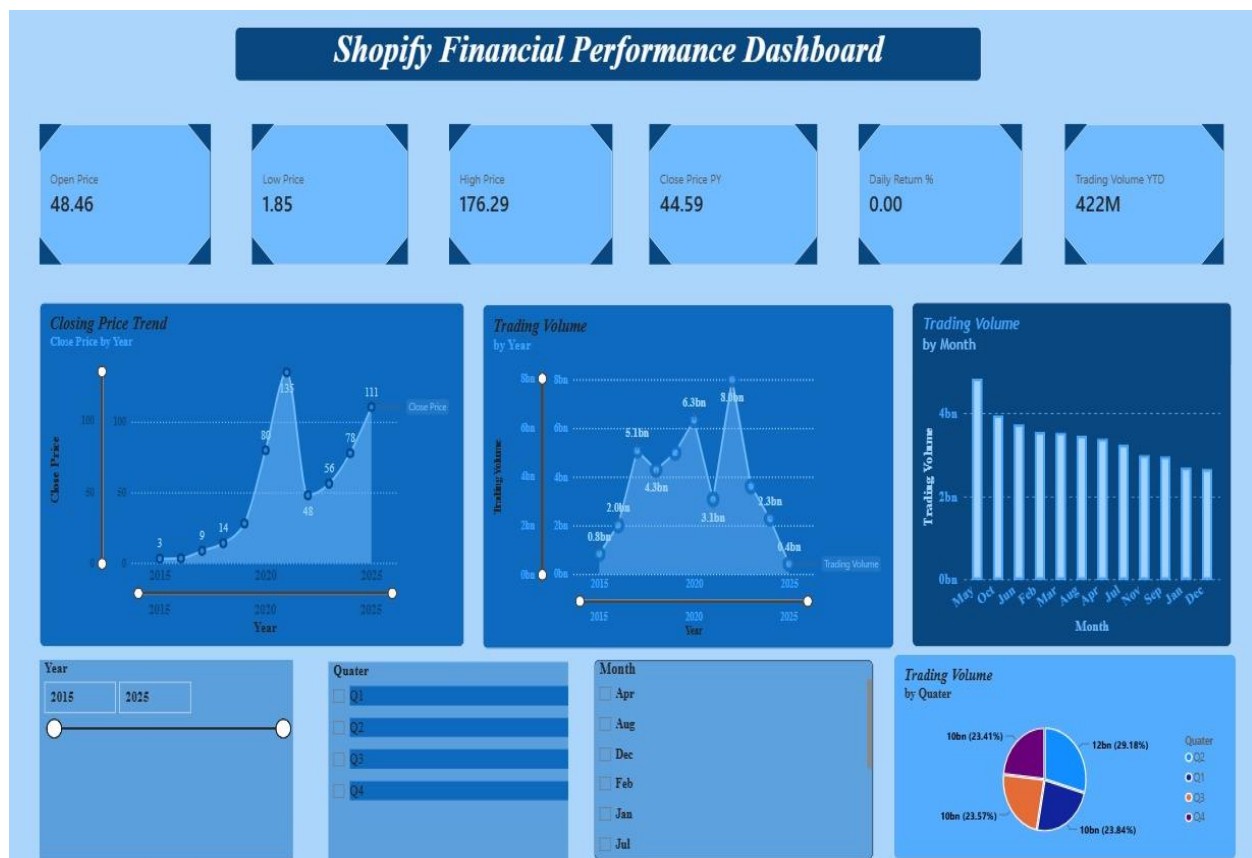
The **Shopify Financial Performance Dashboard** was created using Power BI to analyze Shopify stock performance and trading activity over time. The dashboard provides a clear and interactive view of important financial indicators such as opening price, closing price, high price, low price, daily return percentage, and trading volume.

The dashboard combines **KPI cards, line charts, column charts, pie charts, and slicers** to present the financial information in an easy-to-understand format. Users can interact with the dashboard by selecting different years, quarters, and months.

Dashboard Title

The dashboard is titled “**Shopify Financial Performance Dashboard.**”

The title clearly identifies the purpose of the dashboard and indicates that it focuses on the financial performance of Shopify stock.



KPI Cards

At the top of the dashboard, six KPI cards are provided to display important financial indicators.

- Open Price

The **Open Price** card displays the average opening price of Shopify stock for the selected period. It provides an overview of the stock price at the beginning of the trading period.

- Low Price

The **Low Price** card displays the lowest stock price recorded within the selected period. It helps identify the minimum price level reached by the stock.

- High Price

The **High Price** card displays the highest stock price recorded during the selected period. It helps users understand the maximum price level achieved by Shopify stock.

- Close Price PY

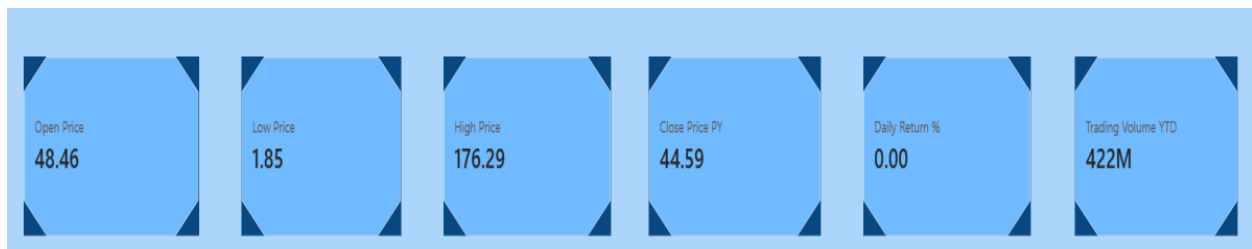
The **Close Price PY** card displays the closing price for the corresponding period in the previous year. This value is calculated using the `SAMEPERIODLASTYEAR()` function and is useful for year-over-year comparison.

- Daily Return %

The **Daily Return %** card displays the percentage change between the current closing price and the previous closing price. It helps users understand the daily movement of the stock.

- Trading Volume YTD

The **Trading Volume YTD** card displays the cumulative trading volume from the beginning of the year up to the selected date. This value is calculated using the `TOTALYTD()` function.



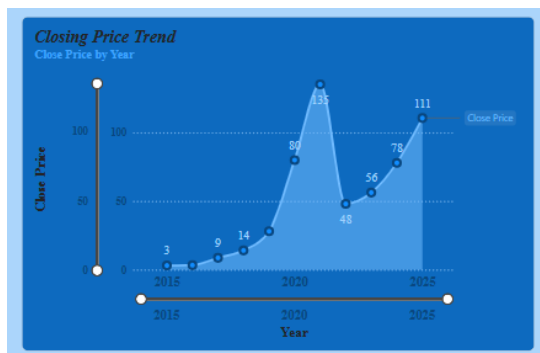
Closing Price Trend

The **Closing Price Trend** line chart shows the movement of Shopify's closing price across different years.

The Year field from the Date Table is used on the horizontal axis, while the Close Price measure is used to display the stock price.

The line chart makes it easy to observe changes in the closing price over time. It also allows users to identify periods where the stock price increased or decreased.

The chart supports time-based analysis and can respond to the selections made using the dashboard slicers.

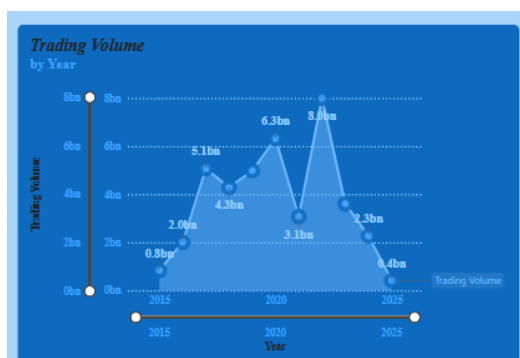


Trading Volume by Year

The **Trading Volume by Year** chart displays the total trading volume for different years.

The Year field is used to group the data, while the Trading Volume measure calculates the total volume.

This visualization helps users compare trading activity across different years. Higher values indicate periods with greater trading activity, while lower values represent relatively lower trading volume.



Trading Volume by Month

The **Trading Volume by Month** column chart shows the trading volume for each month.

The Month field from the Date Table is used to organize the data chronologically, while Trading Volume is used as the measure.

This chart provides a monthly view of trading activity and makes it easier to identify months with relatively higher or lower trading volume.

The Month Number column in the Date Table can also be used to maintain the correct chronological order of the months.



Trading Volume by Quarter

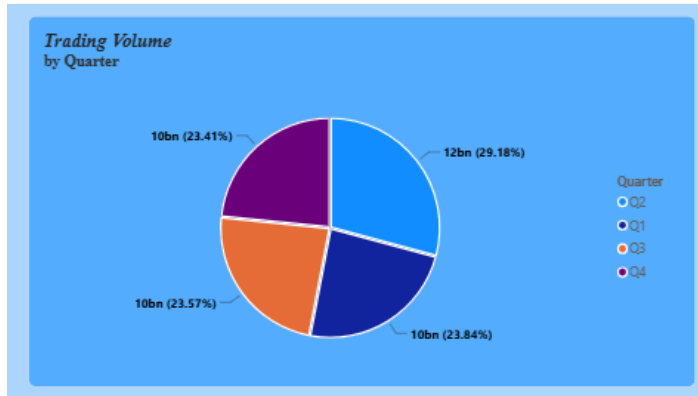
The **Trading Volume by Quarter** pie chart represents the distribution of trading volume across the four quarters.

The four categories are:

- Q1
- Q2
- Q3
- Q4

The chart allows users to compare the contribution of each quarter to the overall trading volume.

The percentage and volume values displayed in the chart provide a quick understanding of how trading activity is distributed across different quarters.



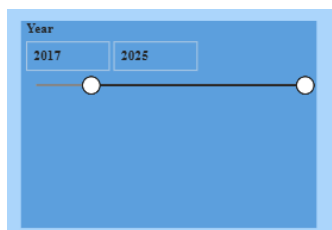
Year Slicer

The **Year slicer** is provided at the bottom-left section of the dashboard.

It allows users to select a particular year or a range of years.

When a year is selected, the other dashboard visuals are automatically filtered according to the selected time period.

This makes it possible to focus the analysis on a specific period.

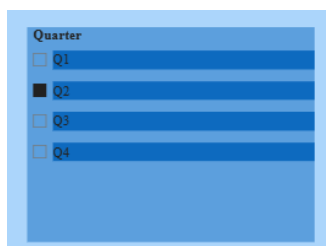


Quarter Slicer

The **Quarter slicer** allows users to filter the dashboard based on Q1, Q2, Q3, or Q4.

For example, selecting Q2 filters the dashboard to show information related to the second quarter.

This provides a more detailed way of analyzing trading activity and stock performance.

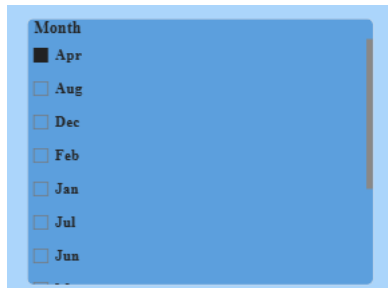


Month Slicer

The **Month slicer** allows users to select individual months.

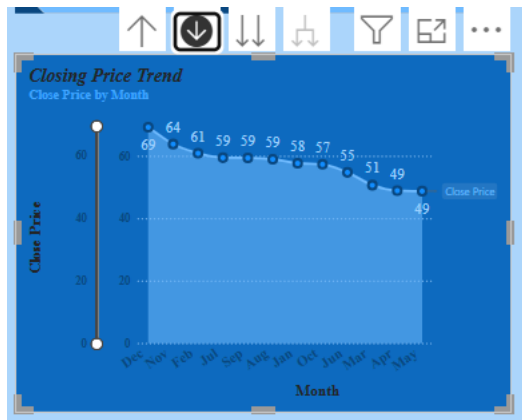
Users can select months such as January, February, April, August, and other available months.

The selected month affects the dashboard visuals and allows users to analyze stock performance and trading activity for a specific month.



Drill-Down: Closing Price by Month

The **Closing Price Trend** visual shows the average closing price of Shopify stock over time. Using the **drill-down feature**, the yearly data is explored further at the **Month** level. This provides a detailed view of how the closing price changes from month to month. It helps identify periods of increase or decrease and understand the overall monthly trend in Shopify's closing price.



Conclusion

The **Shopify Financial Performance Dashboard** was developed using **Power BI** to analyse Shopify's stock prices and trading volume over time. The project involved creating a **Date Table, DAX measures, data relationships, slicers, and drill-down features** for effective time-based analysis.

The dashboard presents important financial information through **KPI cards, closing price trends, and trading volume charts**. The drill-down feature provides a detailed view from **Year to Quarter and Month**, while slicers allow users to filter the data easily.

Overall, the project demonstrates how **Power BI and DAX** can transform raw stock market data into an **interactive, clear, and useful dashboard** for understanding Shopify's financial performance.